

POE-PROG6212

Nokuthula Mnisi
ST10476381 PART 1

SQL Below:

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--ST10476381_NOKUTHULA_MNISI--

--create the Raceday Database if it doesnt exist
IF NOT EXISTS (SELECT name FROM sys.databases WHERE name = 'RaceDay')
BEGIN
    CREATE DATABASE RaceDay;
END
GO

--Switch into the RaceDay Database for everything that follows
USE RaceDay;
GO

-- Drop tables if they already exist (run in reverse dependency order)
IF OBJECT_ID('dbo.RESULT', 'U') IS NOT NULL DROP TABLE dbo.RESULT;
IF OBJECT_ID('dbo.ENROLMENT', 'U') IS NOT NULL DROP TABLE dbo.ENROLMENT;
IF OBJECT_ID('dbo.CATEGORY', 'U') IS NOT NULL DROP TABLE dbo.CATEGORY;
IF OBJECT_ID('dbo.EVENT', 'U') IS NOT NULL DROP TABLE dbo.EVENT;
IF OBJECT_ID('dbo.PARTICIPANT', 'U') IS NOT NULL DROP TABLE dbo.PARTICIPANT;
IF OBJECT_ID('dbo.ORGANISER', 'U') IS NOT NULL DROP TABLE dbo.ORGANISER;
GO

/* Table: ORGANISER */

CREATE TABLE dbo.ORGANISER (
    OrganiserID      INT IDENTITY(1,1) PRIMARY KEY,
    Name             VARCHAR(100)      NOT NULL,
    Email            VARCHAR(150)      NOT NULL UNIQUE,
    Password         VARCHAR(255)      NOT NULL,
    Phone            VARCHAR(20)       NULL
);
GO

/* Table: PARTICIPANT */
CREATE TABLE dbo.PARTICIPANT (
    ParticipantID    INT IDENTITY(1,1) PRIMARY KEY,
    Name             VARCHAR(100)      NOT NULL,
    Email            VARCHAR(150)      NOT NULL UNIQUE,
    Password         VARCHAR(255)      NOT NULL,
    DateOfBirth      DATE              NULL,
    Gender           VARCHAR(20)       NULL
);
GO
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/* Table: EVENT
   FK: OrganiserID -> ORGANISER (1 Organiser : M Events)
*/
CREATE TABLE dbo.EVENT (
    EventID          INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserID      INT                NOT NULL,
    EventName        VARCHAR(150)       NOT NULL,
    EventDate        DATE                NOT NULL,
    Location         VARCHAR(150)       NOT NULL,
    Description      VARCHAR(MAX)       NULL,
    CONSTRAINT FK_Event_Organiser FOREIGN KEY (OrganiserID)
        REFERENCES dbo.ORGANISER(OrganiserID)
);
GO

/* Table: CATEGORY
   FK: EventID -> EVENT (1 Event : M Categories)
*/
CREATE TABLE dbo.CATEGORY (
    CategoryID       INT IDENTITY(1,1) PRIMARY KEY,
    EventID          INT                NOT NULL,
    CategoryName     VARCHAR(100)       NOT NULL,
    Distance         DECIMAL(5,2)       NOT NULL,
    MaxParticipants  INT                NOT NULL DEFAULT 100,
    EntryFee         DECIMAL(8,2)       NOT NULL DEFAULT 0.00,
    CONSTRAINT FK_Category_Event FOREIGN KEY (EventID)
        REFERENCES dbo.EVENT(EventID)
);
GO

/* Table: ENROLMENT
   FK: ParticipantID -> PARTICIPANT (1 Participant : M Enrolments)
   FK: CategoryID -> CATEGORY (1 Category : M Enrolments)
*/
CREATE TABLE dbo.ENROLMENT (
    EnrolmentID      INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID    INT                NOT NULL,
    CategoryID       INT                NOT NULL,
    EnrolmentDate    DATE                NOT NULL DEFAULT CAST(GETDATE() AS DATE),
    Status           VARCHAR(20)        NOT NULL DEFAULT 'Confirmed',
    CONSTRAINT FK_Enrolment_Participant FOREIGN KEY (ParticipantID)
        REFERENCES dbo.PARTICIPANT(ParticipantID),
    CONSTRAINT FK_Enrolment_Category FOREIGN KEY (CategoryID)
        REFERENCES dbo.CATEGORY(CategoryID),
    CONSTRAINT UQ_Enrolment_Participant_Category
        UNIQUE (ParticipantID, CategoryID)
);

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);
GO

/* Table: RESULT
   FK: EnrolmentID -> ENROLMENT (1 Enrolment : 0..1 Result)
   UNIQUE on EnrolmentID enforces the 1-to-0/1 relationship
*/
CREATE TABLE dbo.RESULT (
    ResultID          INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentID       INT                NOT NULL UNIQUE,
    FinishTime        TIME                NULL,
    Position          INT                NULL,
    Status            VARCHAR(20)        NOT NULL DEFAULT 'Pending',
    CONSTRAINT FK_Result_Enrolment FOREIGN KEY (EnrolmentID)
        REFERENCES dbo.ENROLMENT(EnrolmentID)
);
GO

/* seed DATA */

-- Organisers (2)
INSERT INTO dbo.ORGANISER (Name, Email, Password, Phone) VALUES
('Durban Athletics Club', 'events@durbanathletics.co.za', 'HashedPass123',
'0311234567'),
('Cape Peninsula Sports Trust', 'info@cpst.org.za', 'HashedPass456',
'0219876543');

-- Participants (2)
INSERT INTO dbo.PARTICIPANT (Name, Email, Password, DateOfBirth, Gender)
VALUES
('Ntokozo Zulu', 'ntokozo.zulu@email.com', 'HashedPassA1', '1996-04-12',
'Female'),
('Sipho Mooi', 'sipho.mooi@email.com', 'HashedPassB2', '1990-11-03',
'Male');

-- Events (3)
INSERT INTO dbo.EVENT (OrganiserID, EventName, EventDate, Location,
Description) VALUES
(1, 'Durban Beachfront Fun Run', '2026-11-15', 'Durban Beachfront', 'A family-
friendly road running event along the Durban promenade.'),
(1, 'Midlands Trail Challenge', '2026-12-06', 'Pietermaritzburg', 'A scenic
trail running event through the KZN Midlands.'),
(2, 'Cape Town Cycle Classic', '2027-01-24', 'Cape Town CBD', 'An annual
community cycling event around the Cape Peninsula.');

-- Categories (at least one per event)

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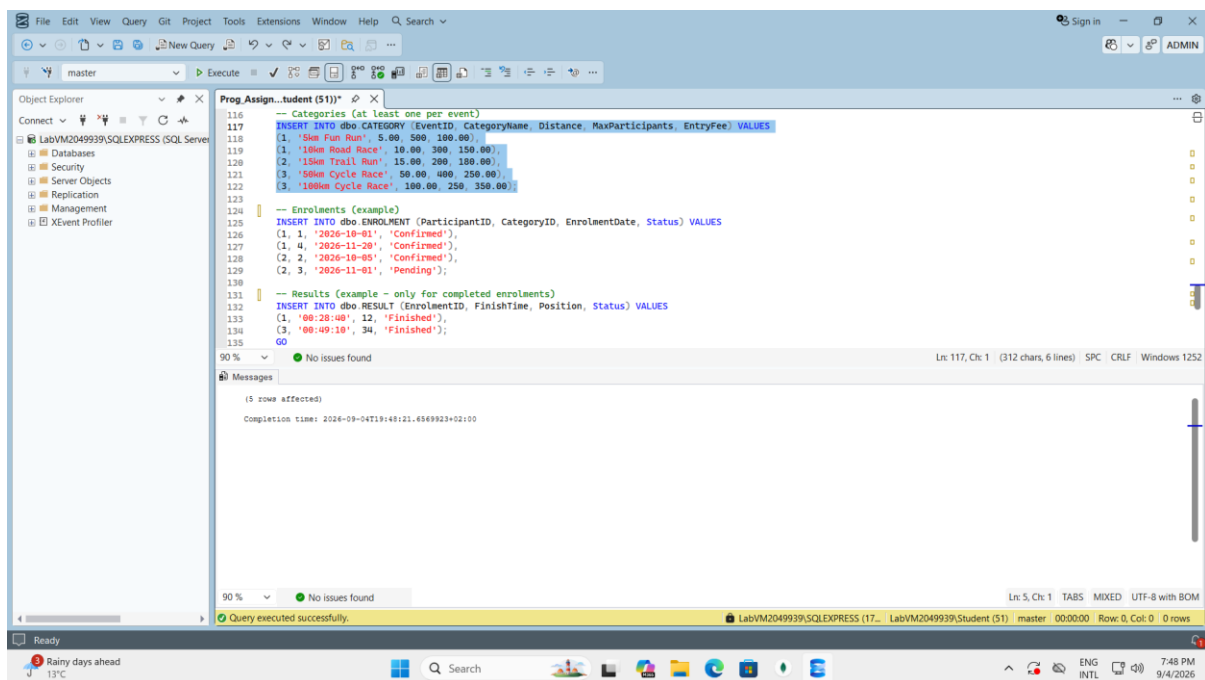
INSERT INTO dbo.CATEGORY (EventID, CategoryName, Distance, MaxParticipants,
EntryFee) VALUES
(1, '5km Fun Run', 5.00, 500, 100.00),
(1, '10km Road Race', 10.00, 300, 150.00),
(2, '15km Trail Run', 15.00, 200, 180.00),
(3, '50km Cycle Race', 50.00, 400, 250.00),
(3, '100km Cycle Race', 100.00, 250, 350.00);

-- Enrolments (example)
INSERT INTO dbo.ENROLMENT (ParticipantID, CategoryID, EnrolmentDate, Status)
VALUES
(1, 1, '2026-10-01', 'Confirmed'),
(1, 4, '2026-11-20', 'Confirmed'),
(2, 2, '2026-10-05', 'Confirmed'),
(2, 3, '2026-11-01', 'Pending');

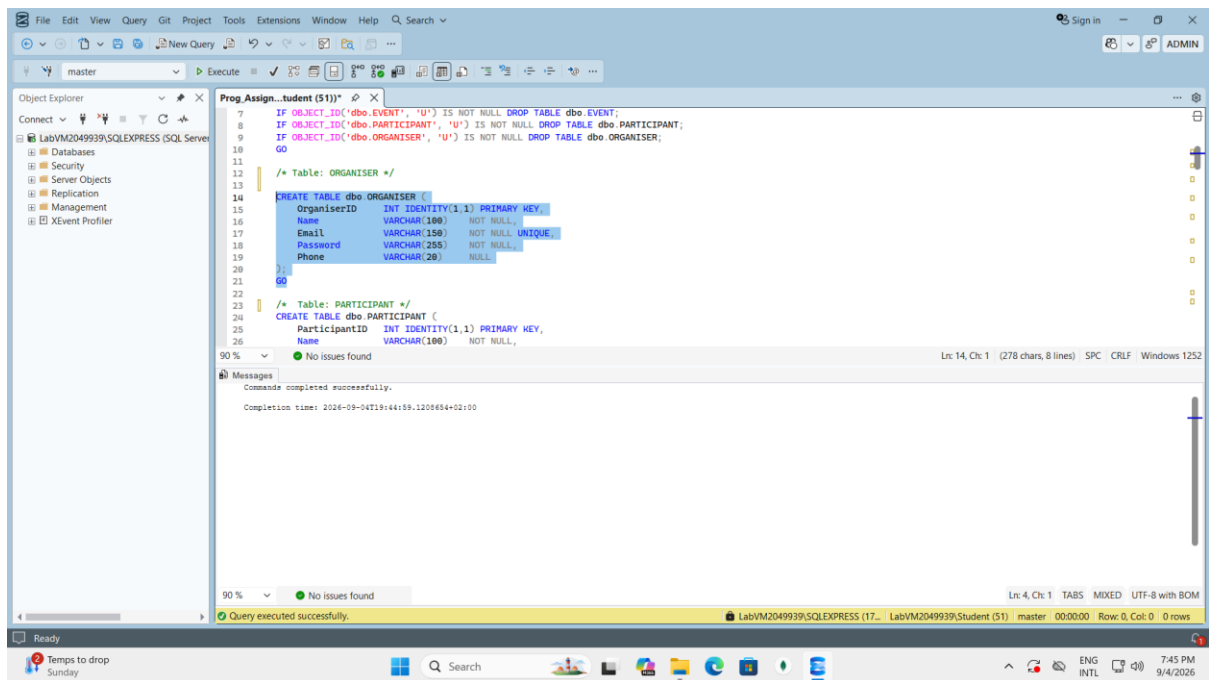
-- Results (example - only for completed enrolments)
INSERT INTO dbo.RESULT (EnrolmentID, FinishTime, Position, Status) VALUES
(1, '00:28:40', 12, 'Finished'),
(3, '00:49:10', 34, 'Finished');
GO

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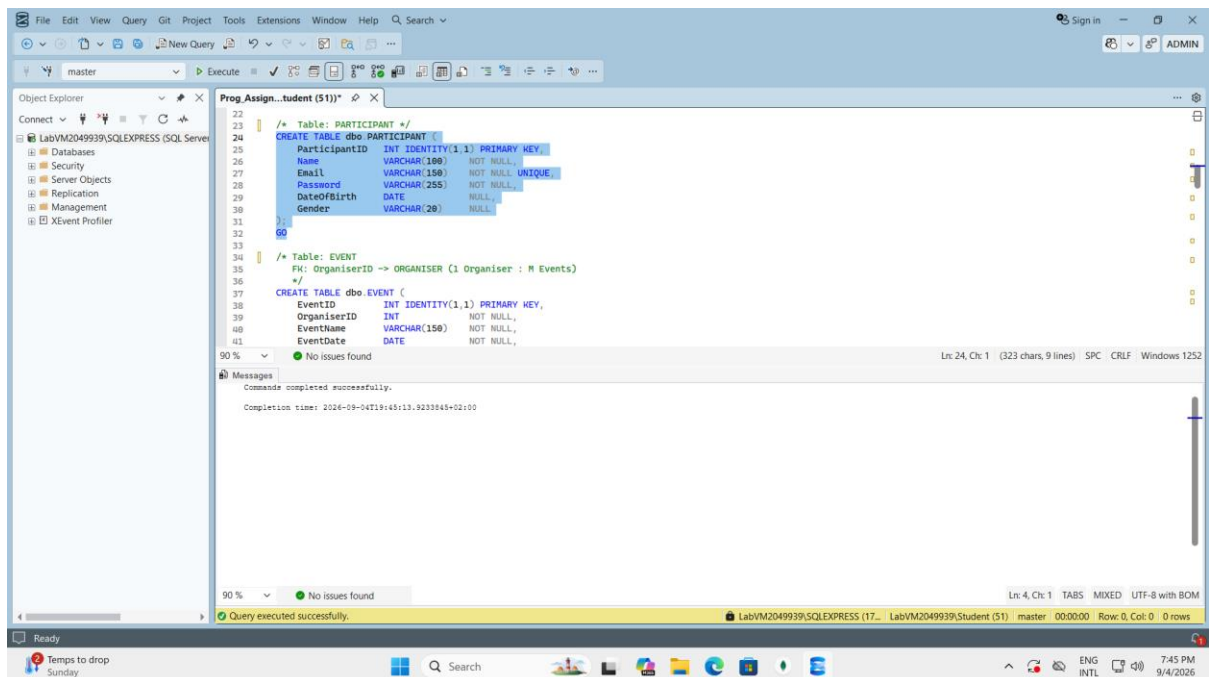
Screenshot of successful executions



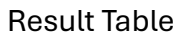
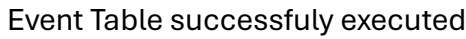
Category table successfully executed

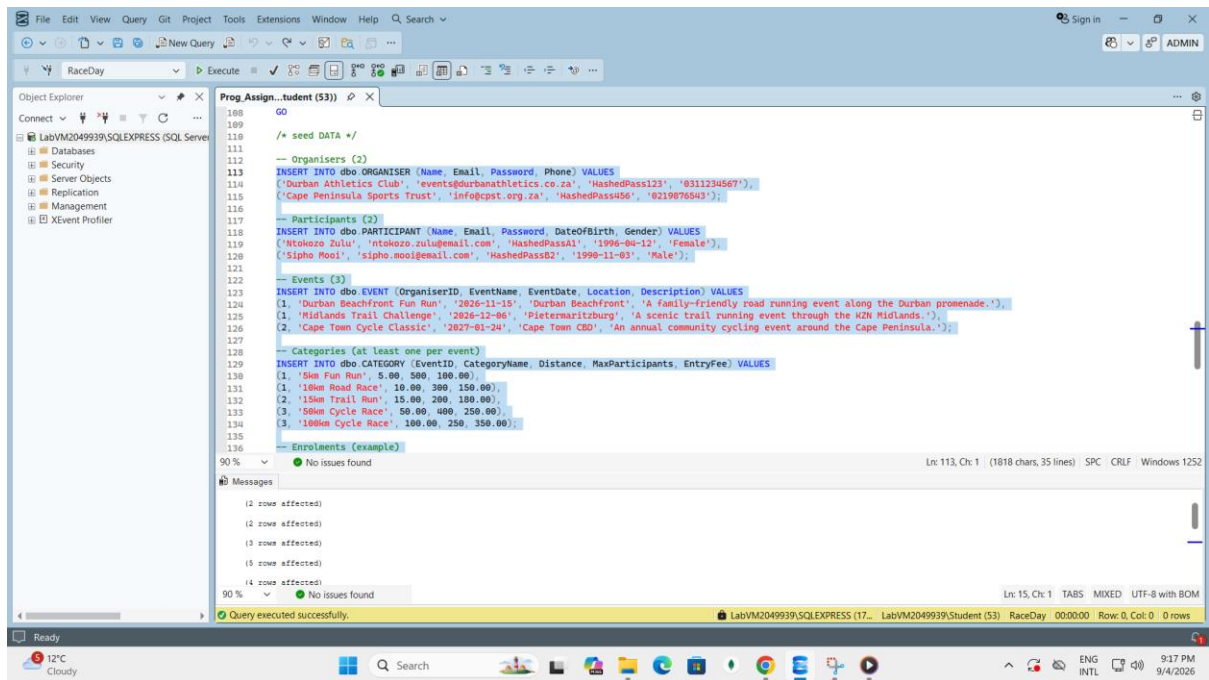


Organiser Table

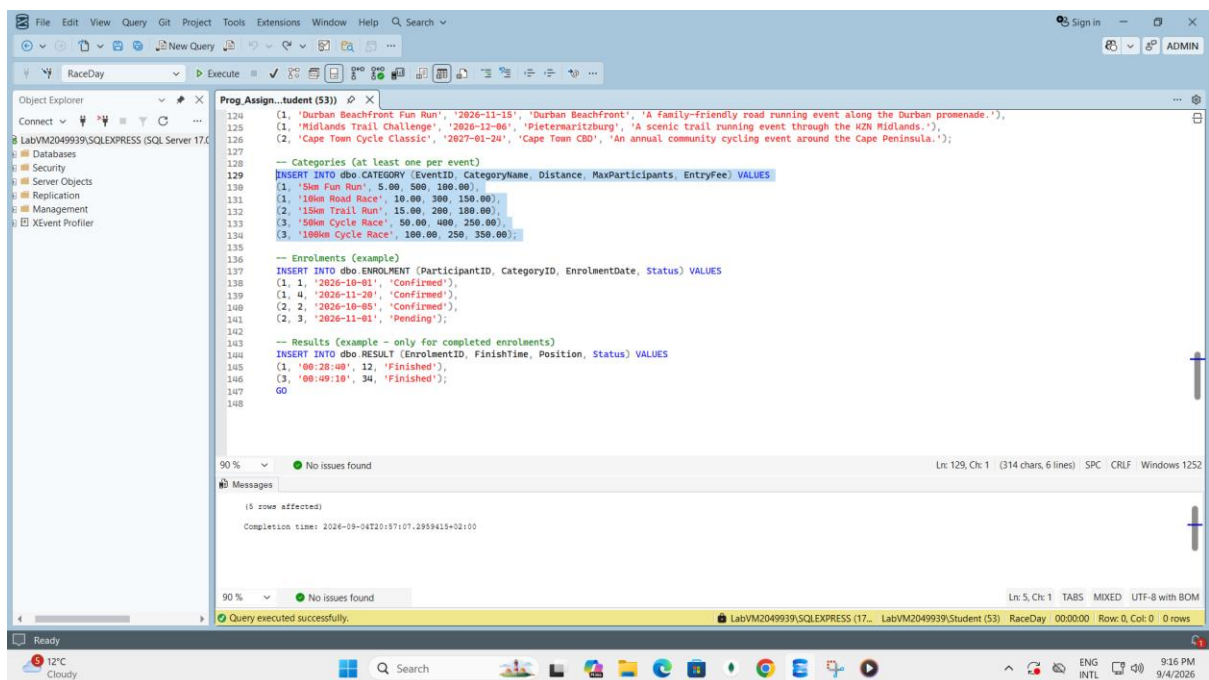


Participant Table seccessfully executed

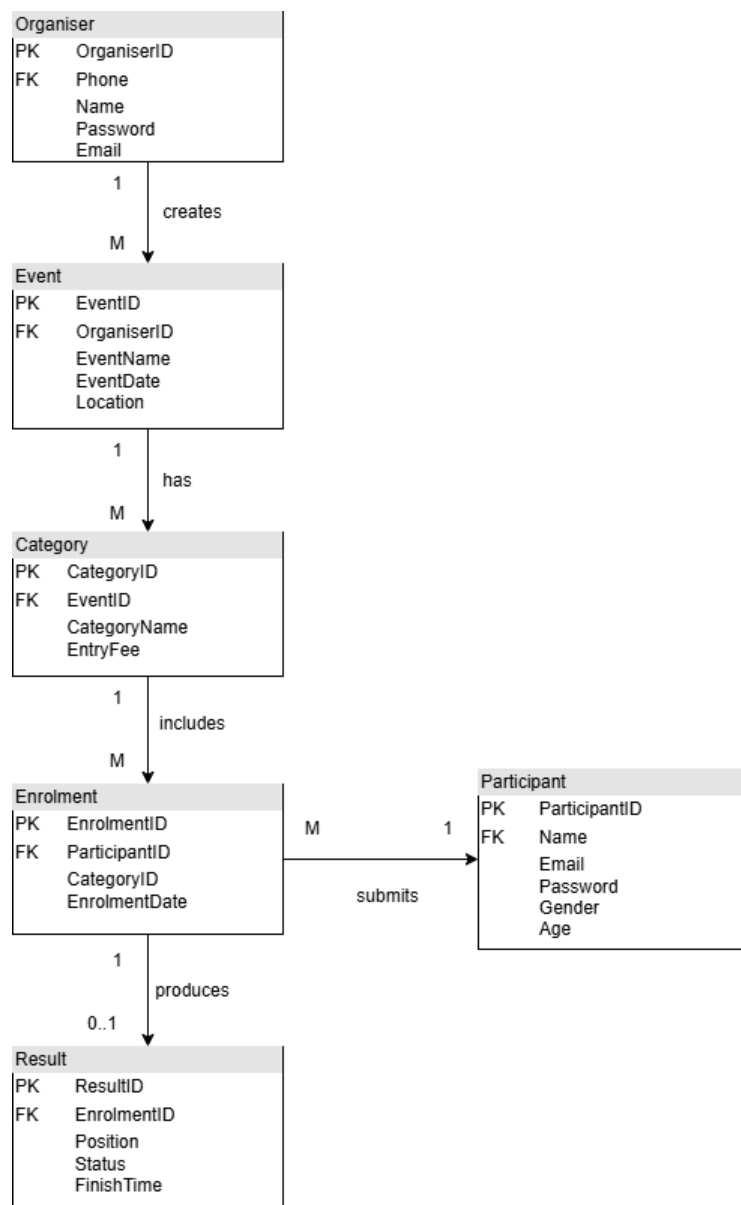




Inserted Values successfully executed



Insert Categories



ERD with 6 entities containing cardinalities for every relationship